

The empirical recurrence for OEIS A302378, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A302378 records “Empirical recurrence of order 84” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 105$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 87$. Its last coefficient is $c_{84} = -42688 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A302378 is “Number of $n \times 5$ 0..1 arrays with every element equal to 1, 2, 3 or 4 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 161, 2886, 56541, 1089035, 20993054, 404225195, 7787623959, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Apr 06 2018 12:40:03, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 105$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 105$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 210$ exact terms from $n = 3$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	13	c_{22}	449597004479	c_{43}	31389069105916	c_{64}	−41890696116
c_2	153	c_{23}	625854728886	c_{44}	−3982605199236	c_{65}	71150266512
c_3	−182	c_{24}	−533977975992	c_{45}	−10508920132044	c_{66}	−91982916576
c_4	−7989	c_{25}	−1367410510405	c_{46}	−18178321397814	c_{67}	−188397110866
c_5	−20241	c_{26}	−924519367769	c_{47}	−20436628098179	c_{68}	−65773075958
c_6	160569	c_{27}	418217406500	c_{48}	−15478133689125	c_{69}	88548514733
c_7	845320	c_{28}	5262379553432	c_{49}	6854927313217	c_{70}	50132007459
c_8	−844551	c_{29}	6534026323621	c_{50}	22023820282739	c_{71}	7811191
c_9	−13873676	c_{30}	−11715614263635	c_{51}	26210069377487	c_{72}	−7033562124
c_{10}	−12913115	c_{31}	−23391057398351	c_{52}	−1992324269808	c_{73}	−9050496368
c_{11}	107580113	c_{32}	8580180578609	c_{53}	−22179981980245	c_{74}	−1133169477
c_{12}	217225463	c_{33}	38444057920376	c_{54}	−3435132598056	c_{75}	1332295044
c_{13}	−449741291	c_{34}	19175270966016	c_{55}	6083653135199	c_{76}	788787258
c_{14}	−1013882920	c_{35}	−23705725307274	c_{56}	7314330879562	c_{77}	54708392
c_{15}	1422571589	c_{36}	−42167923436147	c_{57}	1859533377118	c_{78}	−89703688
c_{16}	−112473516	c_{37}	−5145608850358	c_{58}	−5229467883893	c_{79}	−32725816
c_{17}	−7256089384	c_{38}	29284862676663	c_{59}	−3469508284499	c_{80}	4073128
c_{18}	24310800463	c_{39}	2551231500967	c_{60}	1382645943831	c_{81}	3321216
c_{19}	36316098960	c_{40}	−20422588699483	c_{61}	1289623344040	c_{82}	142256
c_{20}	−147067095941	c_{41}	480338284757	c_{62}	246285072832	c_{83}	−200192
c_{21}	−161124207879	c_{42}	19990792774806	c_{63}	39545342307	c_{84}	−42688

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 87$, and it is the recurrence OEIS A302378 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1 x^{84-1} - \dots - c_{84}$. Its constant term is $-c_{84} = 42688$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -42688 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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